

De facto personal financial archives:

Designing around opportunism, optimism, and benign neglect of personal financial information

Robert D, Ferguson

Research Ireland Centre FutureNeuro, University College Dublin, robert.ferguson@ucd.ie

Banking apps and websites become “de facto financial archives” when used as a long-term strategy to preserve and access personal financial documents and data. Optimism, opportunism, and benign neglect of financial records presents both design challenges and opportunities for the future of HCI and money. This position paper reviews related works on invisible information work, personal archiving burden, and de facto archiving strategies. It then presents examples of de facto financial archiving among young adults (18-25) collected through in-home guided tours. It concludes with a discussion of design implications for the future of HCI and money such as adding additional personal archiving features to enhance recordkeeping functionalities of existing systems.

CCS CONCEPTS • Document management and text processing • User characteristics • Human computer interaction

Additional Keywords and Phrases: Personal information management, Personal archiving, Financial recordkeeping, Young adults, Banking apps and websites, Benign neglect, Optimism, Opportunism

1 INTRODUCTION

De facto archiving entails uncritical reliance on online service providers to maintain and provide long-term access to digital objects often resulting in loss of personal information [16, 18, 19]. Fintech, in the form of banking apps and websites, increasingly becomes de facto financial archives when entrusted to curate our personal financial data and documents. But why do some people trust financial apps and websites to maintain their collections of personal financial information? How can we reimagine the recordkeeping experiences in fintech to ensure access over time? In this position paper, I argue that de facto archiving and recordkeeping burdens are key design challenges for the future of money and HCI. To explore this problem space, I present qualitative data collected from a field study on young adult personal financial information management..

2 RELATED WORKS

Personal information management (PIM) is information work associated with the maintenance of personal collections that are unique, actionable, and informative to individuals [1]. Information work refers to the labor associated with creating, finding, using, and sharing information with a particular emphasis on its costs as time, effort, resources, and outcomes [7]. Due to limited recognition of the labor costs and value generated by PIM, non-professional recordkeeping activities are frequently taken-for-granted by professionals and under-supported by technologies [4, 5, 21]. In the context of personal finance, the frequent roles financial data and records play in managing the financial self, navigating life transitions, and collaborating with others are often invisible and under-valued [6, 11, 12]. Despite the existence of specialized apps, people make limited use of existing financial information management systems, instead relying on analog records and labor-intensive practices to maintain awareness of their current financial situation [15]. As such, adoption of

novel fintech solutions is impacted by the match between a user's information needs and a system's capacity to simplify the management of financial data and documents.

De facto archiving entails a combination of three overlapping beliefs and practices: opportunism, optimism, and benign neglect. *Opportunism* refers to the practice of distributing files over a variety of online services not intended for long-term preservation of digital objects [16, 18, 19]. *Optimism* refers to the belief that digital objects will persist online for long periods of time and are therefore infinitely regenerable [16, 18, 19]. *Benign neglect* refers to a belief that digital objects are inherently ephemeral in nature [16, 18, 19]. In combination, de facto archiving is used to rationalize limited personal investment in digital curation as people cope with the invisible burdens of stewardship [16, 18, 19]. A significant challenge created by de facto archiving is how discovery of loss often occurs after a significant period, thus limiting opportunities for system-enabled recovery [19]. The tendency towards de facto archiving is a specific design opportunity for the future of HCI and money that can be addressed through a more careful consideration of the information and preservation needs presently unfulfilled by existing fintech.

3 EXAMPLES OF FINANCIAL DE FACTO FINANCIAL ARCHIVING AMONG YOUNG ADULTS

To support this argument, I present qualitative data collected from a field study of twenty-three young adults (18-25) in Eastern Canada [11]. During guided tours of personal collections of financial information, participants reflected on their integration of fintech into their recordkeeping practices, which provided insight into the ways in which existing banking apps and websites are frequently treated as de facto archives.

3.1 Optimistic use of fintech to preserve financial records

Optimism about fintech as a long-term strategy was expressed as an uncritical belief that digital documents exist online in perpetuity. For example, Jake, an undergraduate athlete and aspiring actor, described his employer's app that he used to access his pay stubs as, *"a permanent record of what I was paid so I don't ever keep it updated, it already is the way it is."* Similarly, Jenny, an undergraduate economics student, saw her online banking statements as having a diamond-like form of permanence, *"I think they are there forever, right? Because they are all online."* Some participants even expressed the belief that digital documents are impervious to loss like Mike, an international undergraduate student, who said, *"if it is a bill or something, I can always look it up online but I have lost credit cards and stuff before."* Across tours, young adults rarely expressed accurate awareness of service provider retention periods or consideration of the possibility that financial apps and websites could become unavailable in the future for various reasons.

3.2 Opportunistic use of fintech to reduce labour in financial recordkeeping

Several young adults also described opportunistic use of financial apps and websites to reduce physical, cognitive, or affective labours associated with maintaining personal information. For example, Mike valued financial apps and websites for reducing the amount of time and effort spent on managing paper records, *"I like e-statements better because they keep all the records for you. So it is actually much easier. Less paper, especially because I don't know how permanently I am here in Montreal, so to transport all of that stuff is a lot of work."* Similarly, Jake strategically adopted fintech to simplify workflows and limit physical accumulation, *"my bank account information or credit card statements, also, anything like insurance claims are all done online. I don't have anything on paper for that anymore. I used to but as soon as they had the application, I used the application."* But fintech did not always address affective financial labour including motivation. Cathy, a recent graduate working for a game design company, who said, *"I am not really one to keep records. I feel like I try because I know it is something that I should be doing but I consciously don't make an effort to do it."*

4 DESIGN IMPLICATIONS FOR THE FUTURE OF MONEY AND HCI

The combination of optimism and opportunistic use of financial apps and websites to store and provide long-term access to personal financial data and documents results in a form of *benign neglect*, in which curatorial responsibility for records is outsourced by users to fintech. Reliance on fintech to preserve financial records allows for young adults to position themselves as *consumers* rather than *curators* of their financial data and documentation, potentially resulting in loss of control and access over time [1]. Designers should remain ever mindful of Zipf's finding that people always seek to minimize current and future effort when evaluating potential solutions [22]. Therefore, future fintech should: (1) be mindful of the tendency of consumers to neglect financial recordkeeping when technologies provide means for storing and accessing financial records, and (2) seek to reduce physical, cognitive, and affective burdens associated with financial recordkeeping for their clients.

Existing PIM literature provides some insight into ways to reimagine fintech as a more effective long-term storage solution for financial recordkeeping for individuals or collaborative groups. Designers can create information work zones that support different kinds of interactions with personal financial data and documents (e.g., action and communication in the crossroads; financial planning and decision-making in the archives-on-use; and financial self-understanding in the archives), with the ability for users to supplement and contextualize financial data captured by systems using labeling, annotation, and features that allow customization of visual prominence of information [21]. Implementation would enhance the capability of existing systems to "witness" the financial lives of individuals and groups by supporting capture of supplementary evidence about their transactions [2, 8, 13, 14, 20]. Reduced recordkeeping burdens can also create greater user capacity to engage in meta-level PIM activities including organizing, evaluating, sense-making, and sharing personal information [10], which can enhance the collaborative work and communication around personal finance.

On the flipside, fintech can also be reimaged to respond to de facto archiving by creating opportunities for higher levels of personal recordkeeping accountability and reducing the embeddedness of archiving in systems designed primarily for communication between financial service providers and their clients [9]. Shifting primary responsibility for preservation back onto users may prevent loss of documents as business-as-usual (e.g., from standard retention periods or discontinuation of services within a jurisdiction) [19]. For example, interfaces could be designed to proactively guide users through the creation and execution of data curation plans and support the exportation data or documents in batches at regular intervals with minimal effort [17]. Fintech providers could develop resources that improve recordkeeping literacy among their users and promote critical evaluation and reduce reliance on websites and apps as a preservation strategy for financial data and documents [3].

5 CONCLUSION

When users rely on financial apps and websites to maintain and provide long term access to personal financial data and documents, fintech becomes de facto personal financial archives. Users rely on financial apps and websites to address unresolved labour costs associated with personal financial information management. The physical, cognitive, and affective burdens of PIM represents both a challenge and opportunity for the future of fintech. Optimistic and opportunistic use of financial apps and websites that results in benign neglect can be addressed through intentional rethinking of the recordkeeping functions and experiences built into fintech.

ACKNOWLEDGMENTS

This work was funded by the Social Science and Humanities Research Council of Canada, grant no.: 752-2014-1517. This publication has emanated from research supported in part by a research grant from Science Foundation Ireland (SFI) under Grant Number 21/RC/10294_P2 and co-funded under the European

REFERENCES

- [1] Ofer Bergman and Steve Whittaker. 2016. *The science of managing our digital stuff*. MIT Press, Cambridge, MA.
- [2] Richard J Cox. 1996. The record in the manuscript collection. *Archives and manuscripts* 24, 1 (1996), 46–61.
- [3] Amber L. Cushing. 2010. Highlighting the archives perspective in the personal digital archiving discussion. *Library Hi Tech* 28, 2 (2010), 301–312.
- [4] Amber L. Cushing. 2023. PIM as a caring: Using ethics of care to explore personal information management as a caring process. *Asso for Info Science & Tech* 74, 11 (November 2023), 1282–1292. <https://doi.org/10.1002/asi.24824>
- [5] Amber L. Cushing and Páraic Kerrigan. 2022. Personal information management burden: A framework for describing nonwork personal information management in the context of inequality. *Asso for Info Science & Tech* 73, 11 (November 2022), 1543–1558. <https://doi.org/10.1002/asi.24692>
- [6] Amber L. Cushing, Vanessa Reyes, and Robert D. Ferguson. 2023. Conversations on the Invisibility of PIM , Approaching Themes, and New Avenues of Research. *Proceedings of the Association for Information Science and Technology* 60, 1 (October 2023), 553–559. <https://doi.org/10.1002/pra2.814>
- [7] Nicole K. Dalmer and Isto Huvila. 2019. Conceptualizing information work for health contexts in Library and Information Science. *JD* 76, 1 (October 2019), 96–108. <https://doi.org/10.1108/JD-03-2019-0055>
- [8] Jennifer Douglas. 2015. The archiving “I”: a closer look in the archives of writers. *Archivaria* (2015), 53–89.
- [9] Nicolas Ducheneaut and Victoria Bellotti. 2001. E-mail as habitat: an exploration of embedded personal information management. *interactions* 8, 5 (September 2001), 30–38. <https://doi.org/10.1145/382899.383305>
- [10] Yuanyuan Feng and Denise E. Agosto. 2019. Revisiting personal information management through information practices with activity tracking technology. *Journal of the Association for Information Science and Technology* 70, 12 (December 2019), 1352–1367. <https://doi.org/10.1002/asi.24253>
- [11] Robert D Ferguson. 2023. Exploring the personal financial information management of young adults. McGill University, Montreal, Canada. Retrieved from <https://escholarship.mcgill.ca/concern/theses/kw52jf25g>
- [12] Robert D Ferguson, Karyn Moffatt, and Rebekah Willson. Forthcoming. “Am I being responsible?": Navigating coming-of-age transitions through personal financial information management (PFIM). *Journal of the Association for Information Science and Technology* (Forthcoming). <https://doi.org/10.1002/asi.24990>
- [13] Catherine Hobbs. 2001. The character of personal archives: Reflections on the value of records of individuals. *Archivaria* (2001), 126–135.
- [14] Catherine Hobbs. 2017. Reenvisioning the Personal: Reframing Traces of Individual Life. In *Currents of archival thinking* (Second edition), Heather MacNeil and Terry Eastwood (eds.). Libraries Unlimited, an imprint of ABC-CLIO, LLC, Santa Barbara, California Denver, Colorado, 213–241.
- [15] Joseph Jofish Kaye, Mary McCuiston, Rebecca Gulotta, and David A. Shamma. 2014. Money talks: tracking personal finances. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, April 26, 2014. ACM, Toronto Ontario Canada, 521–530. <https://doi.org/10.1145/2556288.2556975>
- [16] Catherine C. Marshall. 2008. Rethinking Personal Digital Archiving, Part 1: Four Challenges from the Field. *D-Lib Magazine* 14, 3/4 (March 2008). <https://doi.org/10.1045/march2008-marshall-pt1>
- [17] Catherine C. Marshall. 2008. Rethinking Personal Digital Archiving, Part 2: Implications for Services, Applications, and Institutions. *D-Lib Magazine* 14, 3/4 (March 2008). <https://doi.org/10.1045/march2008-marshall-pt2>

- [18] Catherine C Marshall, Sara Bly, and Francoise Brun-Cottan. 2007. The long term fate of our digital belongings: Toward a service model for personal archives. *arXiv preprint arXiv:0704.3653* (2007).
- [19] Catherine C. Marshall, Frank McCown, and Michael L. Nelson. 2007. Evaluating Personal Archiving Strategies for Internet-based Information. (2007). <https://doi.org/10.48550/ARXIV.0704.3647>
- [20] Sue McKemmish. 1996. Evidence of *me*. *The Australian Library Journal* 45, 3 (January 1996), 174–187. <https://doi.org/10.1080/00049670.1996.10755757>
- [21] Enrico Maria Piras and Alberto Zanutto. 2010. Prescriptions, X-rays and Grocery Lists. Designing a Personal Health Record to Support (The Invisible Work Of) Health Information Management in the Household. *Comp Supported Coop Work* 19, 6 (December 2010), 585–613. <https://doi.org/10.1007/s10606-010-9128-5>
- [22] George Kingsley Zipf. 2012. *Human behavior and the principle of least effort: an introduction to human ecology*. Martino Publishing [u.a.], Mansfield Centre, Conn.